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1  ----- HR Analysis Project 25+ insights -----
2  Use AdventureWorks2022;
3
4  -- 1. Total Employees by Gender
5  Select Gender , COUNT(*) As TotalEmployees
6  From HumanResources.Employee
7  Group by Gender;
8
9  -- 2. Employees Count by Marital Status
10 Select MaritalStatus, COUNT(*) As EmployeeCount
11 from HumanResources.Employee
12 Group by MaritalStatus;
13
14 -- 3. Employee Age Distribution
15 Select DATEPART(YEAR, GETDATE()) - Year(BirthDate) As Age, Count(*) As Count
16 From HumanResources.Employee
17 Group by DATEPART(YEAR, GETDATE())- Year(BirthDate)
18 Order by Age;
19
20 -- 4. Headcount by Year of Hire
21 Select Year(HireDate) as HireYear, Count(*) AS NumberHired
22 From HumanResources.Employee
23 Group by Year(HireDate)
24 Order by HireYear;
25
26 -- 5. Salaried vs Hourly Employees
27 Select
28     Case
29         When SalariedFlag = 1 Then 'Salaried' Else 'Hourly'
30     End As EmployeeType
31     , Count(*) AS Count
32 From HumanResources.Employee e
33 Group by e.SalariedFlag;
34
35 -- 6. Active vs Inactive Employees
36 Select
37     Case
38         When CurrentFlag = 1 Then 'Active' Else 'Inactive'
39     End As Status,
40     Count(*) as Count
41 from HumanResources.Employee e
42 Group by e.CurrentFlag;
43 ----- Alternate -----
44 SELECT
45     SUM(CASE WHEN CurrentFlag = 1 THEN 1 ELSE 0 END) AS Active_Employee,
46     SUM(CASE WHEN CurrentFlag = 0 THEN 1 ELSE 0 END) AS Inactive_Employee
47 FROM HumanResources.Employee e;
48
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49 -- 7. Employee Count by Job Title
50 Select JobTitle, Count(*) as NumberOfEmployee
51 From HumanResources.Employee e
52 Join HumanResources.EmployeeDepartmentHistory edh
53     On e.BusinessEntityID = edh.BusinessEntityID
54 Group by JobTitle
55 Order by NumberOfEmployee desc;
56
57 -- 8. Employee Distribution by Department
58 Select d.Name As Department_Name , Count(*) As EmployeeCount
59 from HumanResources.EmployeeDepartmentHistory edh
60 join HumanResources.Department d
61     On edh.DepartmentID = d.DepartmentID
62 Group by d.Name
63 Order by EmployeeCount desc;
64
65 -- 9. Employee Tenure Buckets
66 Select
67     Case
68         When DATEDIFF(YEAR, HireDate, '2013-05-30') < 1 Then '< 1 year'
69         When DATEDIFF(Year, HireDate, '2013-05-30') Between 1 And 3 Then '1-3 ↗
70             years'
71         When DATEDIFF(Year, HireDate, '2013-05-30') Between 4 And 6 Then '4-6 ↗
72             years'
73         Else '7+ years'
74     End As TenureBucket,
75     Count(*) As NumberOfEmployees
76 from HumanResources.Employee
77 Group by
78     Case
79         When DATEDIFF(YEAR, HireDate, '2013-05-30') < 1 Then '< 1 year'
80         When DATEDIFF(Year, HireDate, '2013-05-30') Between 1 And 3 Then '1-3 ↗
81             years'
82         When DATEDIFF(Year, HireDate, '2013-05-30') Between 4 And 6 Then '4-6 ↗
83             years'
84         Else '7+ years'
85     End
86 Order by TenureBucket;
87
88 -- 10. Monthly Hiring Trend And Quarter Also
89 Select FORMAT(HireDate, 'yyyy-MM') as HireMonth, COUNT(*) as Hires
90 from HumanResources.Employee
91 Group by FORMAT(HireDate, 'yyyy-MM')
92 order by HireMonth;
93
94 Select CAST(YEAR(HireDate) as varchar(4))+ ' -Q'+
95     Cast(DATEPART(QUARTER,HireDate) as varchar(1)) as HireQuarter,
96     Count(*) as Hires
97 from HumanResources.Employee
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94 Group by YEAR(HireDate), DATEPART(QUARTER,HireDate)
95 Order by YEAR(HireDate), DATEPART(QUARTER,HireDate);
96
97
98 --11. Gender Diversity by Department
99 Select d.Name As DepartmentName, e.Gender , Count(*) As HeadCount
100 From HumanResources.Employee e
101 Join HumanResources.EmployeeDepartmentHistory edh
102     On e.BusinessEntityID = edh.BusinessEntityID
103 Join HumanResources.Department d
104     On edh.DepartmentID = d.DepartmentID
105 Group by d.Name, e.Gender
106 Order by DepartmentName, e.Gender;
107
108 -- 12. Total Employees by Job Title and Department
109 Select d.Name As DepartmentName, e.JobTitle , Count(*) As HeadCount
110 From HumanResources.Employee e
111 Join HumanResources.EmployeeDepartmentHistory edh
112     On e.BusinessEntityID = edh.BusinessEntityID
113 Join HumanResources.Department d
114     On edh.DepartmentID = d.DepartmentID
115 Group by d.Name, e.JobTitle
116 Order by DepartmentName, HeadCount desc;
117
118 -- 13. Employee Count by Shift
119 Select s.Name as Shift, COUNT(distinct edh.BusinessEntityID) as EmployeeCount
120 From HumanResources.EmployeeDepartmentHistory edh
121 Join HumanResources.Shift s
122     On edh.ShiftID = s.ShiftID
123 Group by s.Name;
124
125 --14. Department-wise Average Tenure (Years)
126 SELECT
127     d.Name AS Department,
128     AVG(DATEDIFF(YEAR, e.HireDate, '2013-05-30')) AS AvgTenureYears
129 FROM HumanResources.Employee e
130 JOIN HumanResources.EmployeeDepartmentHistory edh
131     ON e.BusinessEntityID = edh.BusinessEntityID
132 JOIN HumanResources.Department d
133     ON edh.DepartmentID = d.DepartmentID
134 WHERE edh.EndDate IS NULL -- sirf active employees ke liye
135 GROUP BY d.Name
136 ORDER BY AvgTenureYears DESC;
137
138 -- 15. Monthly Attrition Trend (Terminated Employees Per Month)
139 SELECT FORMAT(eh.EndDate, 'yyyy-MM') AS TerminationMonth, COUNT(*) AS
    EmployeesTerminated
140 FROM HumanResources.EmployeeDepartmentHistory eh
141 WHERE eh.EndDate IS NOT NULL
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142 GROUP BY FORMAT(eh.EndDate, 'yyyy-MM')
143 ORDER BY TerminationMonth;
144
145 -- 16. Employees by Pay Frequency
146 Select PayFrequency, Count(*) as HeadCount
147 From HumanResources.EmployeePayHistory
148 Group by PayFrequency;
149
150 -- 17. Highest Paid Employee By Department
151 Select d.Name As Department, e.BusinessEntityID, MAX(eph.Rate) as HighstPay
152 from HumanResources.EmployeePayHistory eph
153 Join HumanResources.Employee e          On eph.BusinessEntityID =
    e.BusinessEntityID
154 Join HumanResources.EmployeeDepartmentHistory edh On e.BusinessEntityID =
    edh.BusinessEntityID
155 Join HumanResources.Department d          On edh.DepartmentID = d.DepartmentID
156 Group by d.Name, e.BusinessEntityID;
157
158 -- 18. Employees Promoted More Than Once
159 Select BusinessEntityID, COUNT(Distinct ShiftID) as Promotion
160 from HumanResources.EmployeeDepartmentHistory
161 group by BusinessEntityID
162 Having COUNT(Distinct ShiftID) > 1;
163
164 -- 19. Employees With Most Job Title Changes
165 Select edh.BusinessEntityID, COUNT(Distinct e.Jobtitle) as JobtitleChange
166 from HumanResources.EmployeeDepartmentHistory edh
167 Join HumanResources.Employee e
168     On edh.BusinessEntityID = e.BusinessEntityID
169 Group by edh.BusinessEntityID
170 order by JobtitleChange desc;
171
172 -- 20. Employees Earning More Than Department Average
173 SELECT eph.BusinessEntityID, d.Name AS Department, eph.Rate AS Salary
174 FROM HumanResources.EmployeePayHistory eph
175 JOIN HumanResources.EmployeeDepartmentHistory edh ON eph.BusinessEntityID =
    edh.BusinessEntityID
176 JOIN HumanResources.Department d ON edh.DepartmentID = d.DepartmentID
177 WHERE eph.Rate > (
178     SELECT AVG(eph2.Rate)
179     FROM HumanResources.EmployeePayHistory eph2
180     JOIN HumanResources.EmployeeDepartmentHistory edh2 ON
    eph2.BusinessEntityID = edh2.BusinessEntityID
181     WHERE edh2.DepartmentID = edh.DepartmentID
182 );
183
184 -- 21. Employees jinhe ek se zyada promotions mile
185 SELECT BusinessEntityID, COUNT(DISTINCT ShiftID) AS PromotionCount
186 FROM HumanResources.EmployeeDepartmentHistory
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187 GROUP BY BusinessEntityID
188 HAVING COUNT(DISTINCT ShiftID) > 1;
189
190 -- 22. Employees with no salary hike/raise
191 SELECT BusinessEntityID
192 FROM HumanResources.EmployeePayHistory
193 GROUP BY BusinessEntityID
194 HAVING COUNT(*) = 1;
195
196 -- 23. Department-wise average pay
197 SELECT d.Name AS Department, AVG(eph.Rate) AS AvgPay
198 FROM HumanResources.EmployeePayHistory eph
199 JOIN HumanResources.EmployeeDepartmentHistory edh ON eph.BusinessEntityID =
    edh.BusinessEntityID
200 JOIN HumanResources.Department d ON edh.DepartmentID = d.DepartmentID
201 GROUP BY d.Name;
202
203 -- 24. Highest Tenure Employee(s) in Company
204 SELECT TOP 1 BusinessEntityID, DATEDIFF(year, HireDate, '2013-05-30') AS
    Tenure
205 FROM HumanResources.Employee
206 ORDER BY Tenure DESC;
207
208 -- 25. Employee Rank by Salary Within Department
209 Select
210     e.BusinessEntityID,
211     d.Name As Department,
212     eph.Rate as Salary,
213     RANK() Over (Partition by d.Name Order by eph.Rate Desc) as
        SalaryRankInDept
214 from HumanResources.Employee e
215 Join HumanResources.EmployeeDepartmentHistory edh On e.BusinessEntityID
    = edh.BusinessEntityID
216 Join HumanResources.EmployeePayHistory eph On
    edh.BusinessEntityID = eph.BusinessEntityID
217 Join HumanResources.Department d On edh.DepartmentID =
    d.DepartmentID
218 Where eph.Rate Is Not Null;
219
220 -- 26. Average, Min, Max Tenure by Department
221
222 WITH EmpTenure AS (
223     SELECT
224         e.BusinessEntityID,
225         d.Name AS Department,
226         DATEDIFF(YEAR, e.HireDate, '2013-05-30') AS Tenure
227 FROM HumanResources.Employee e
228 JOIN HumanResources.EmployeeDepartmentHistory edh ON e.BusinessEntityID =
    edh.BusinessEntityID

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229 JOIN HumanResources.Department d ON edh.DepartmentID = d.DepartmentID
230 )
231 SELECT
232     Department,
233     Tenure,
234     AVG(Tenure) OVER (PARTITION BY Department) AS AvgTenure,
235     MIN(Tenure) OVER (PARTITION BY Department) AS MinTenure,
236     MAX(Tenure) OVER (PARTITION BY Department) AS MaxTenure
237 FROM EmpTenure
238 GROUP BY Department, Tenure;
239
240 -- 27. New Hires with Running Total by Month
241 WITH MonthlyHires AS (
242     SELECT
243         FORMAT(HireDate, 'yyyy-MM') AS HireMonth,
244         COUNT(*) AS Hires
245     FROM HumanResources.Employee
246     WHERE HireDate >= DATEADD(YEAR, -4, '2013-05-30')
247     GROUP BY FORMAT(HireDate, 'yyyy-MM')
248 )
249 SELECT
250     HireMonth,
251     Hires,
252     SUM(Hires) OVER (ORDER BY HireMonth ROWS BETWEEN UNBOUNDED PRECEDING AND
253         CURRENT ROW) AS RunningTotal
254 FROM MonthlyHires
255 ORDER BY HireMonth
256
257 -- 28. Previous and Next Job Title Change Per Employee (LEAD/LAG)
258 WITH EmpJobHistory AS (
259     SELECT
260         e.BusinessEntityID,
261         e.JobTitle,
262         edh.StartDate
263     FROM HumanResources.Employee e
264     JOIN HumanResources.EmployeeDepartmentHistory edh ON e.BusinessEntityID =
265         edh.BusinessEntityID
266 )
267 SELECT
268     BusinessEntityID,
269     JobTitle,
270     StartDate,
271     LAG(JobTitle,1) OVER (PARTITION BY BusinessEntityID ORDER BY StartDate) AS
272         PrevJobTitle,
273     LEAD(JobTitle,1) OVER (PARTITION BY BusinessEntityID ORDER BY StartDate) AS
274         NextJobTitle
275 FROM EmpJobHistory
276 ORDER BY BusinessEntityID, StartDate;
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274 -- 29. Find Second Highest Salary by Department (DENSE_RANK)
275 SELECT DISTINCT Department, Salary
276 FROM (
277     SELECT d.Name AS Department, eph.Rate AS Salary,
278            DENSE_RANK() OVER (PARTITION BY d.Name ORDER BY eph.Rate DESC) AS SalaryRank
279     FROM HumanResources.EmployeePayHistory eph
280     JOIN HumanResources.EmployeeDepartmentHistory edh ON eph.BusinessEntityID =
281            edh.BusinessEntityID
282     JOIN HumanResources.Department d ON edh.DepartmentID = d.DepartmentID
283 ) AS RankedSalaries
284 WHERE SalaryRank = 2;
285
286 -- 30
287 WITH MonthlyHires AS (
288     SELECT
289         FORMAT(HireDate, 'yyyy-MM') AS HireMonth,
290         COUNT(*) AS Hires
291     FROM HumanResources.Employee
292     GROUP BY FORMAT(HireDate, 'yyyy-MM')
293 )
294 SELECT
295     HireMonth,
296     Hires,
297     Convert(decimal,AVG(Hires) OVER (ORDER BY HireMonth ROWS BETWEEN 11
298            PRECEDING AND CURRENT ROW)) AS Rolling12MoAvgHires
299 FROM MonthlyHires
300 ORDER BY HireMonth;
301
302
303
304
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